

Design of SR Latch, D Flip-Flop, JK Flip-Flop and T Flip-Flop

Introduction: Sequential Logic and Memory

Sequential circuit
Output depends on the present inputs and the stored past state.

Memory element
Stores one bit (0 or 1) and keeps it until it is changed.

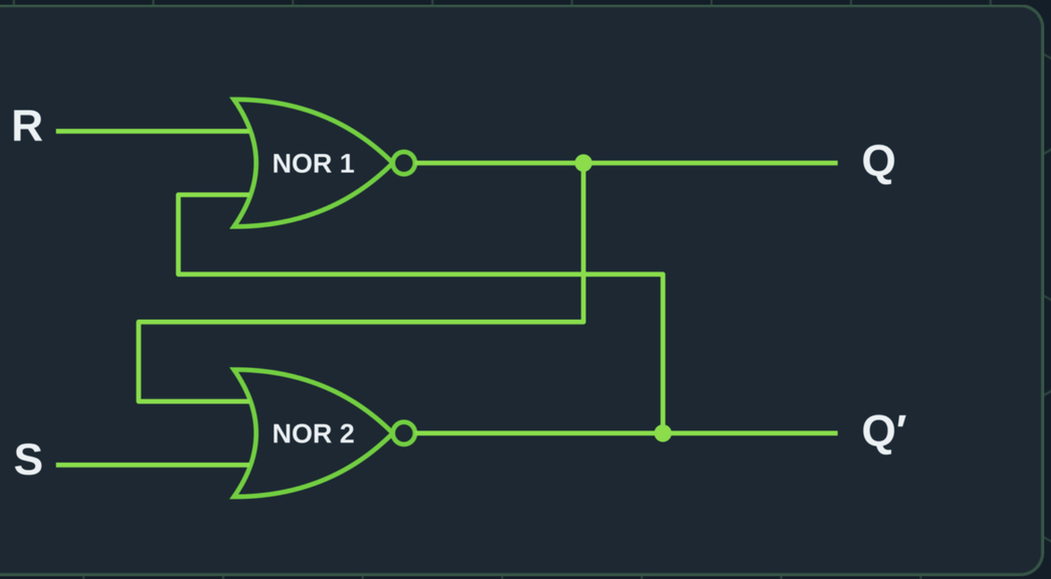
Latch
Level-sensitive memory. Output can change while the enable is active.

Flip-flop
Clocked memory. State changes at a clock edge, once per clock event.

Circuit type	Output depends on
Combinational	Present inputs only: Y = f(inputs)
Sequential	Inputs and present state: Q(n+1) = f(inputs, Qn)

Latches and flip-flops are the basic building blocks that give digital systems memory.

SR Latch Design



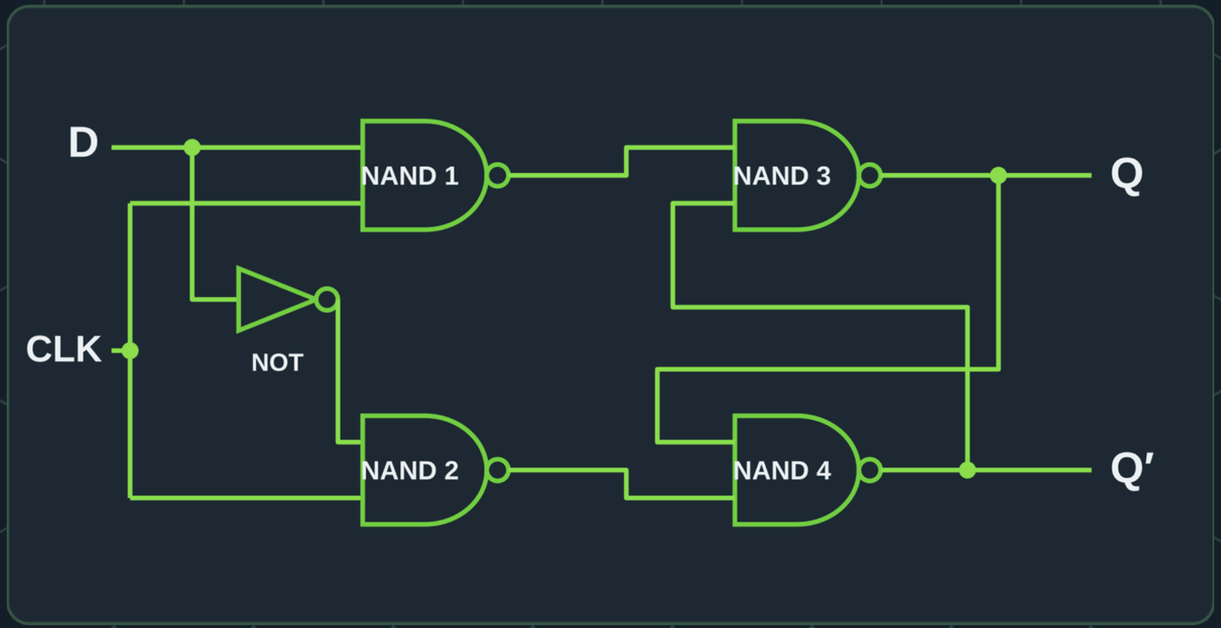
Cross-coupled NOR gates: Q = (R + Q′)′ and Q′ = (S + Q)′

- Each output feeds the other gate, so the pair remembers its state.
- S = 1 sets Q to 1. R = 1 resets Q to 0. S = R = 0 holds.

S	R	Q(n+1)	Operation
0	0	Qn	Hold
0	1	0	Reset
1	0	1	Set
1	1	Invalid	Q = Q′ = 0

Invalid condition: S = R = 1. Both outputs go to 0, so Q and Q′ are no longer complements. If S and R then return to 0 together, the final state is unpredictable.

D Flip-Flop Design



NAND 1 and NAND 2 gate D and D' with CLK. NAND 3 and NAND 4 form the storing latch. Two such latches in master–slave form make an edge-triggered D flip-flop.

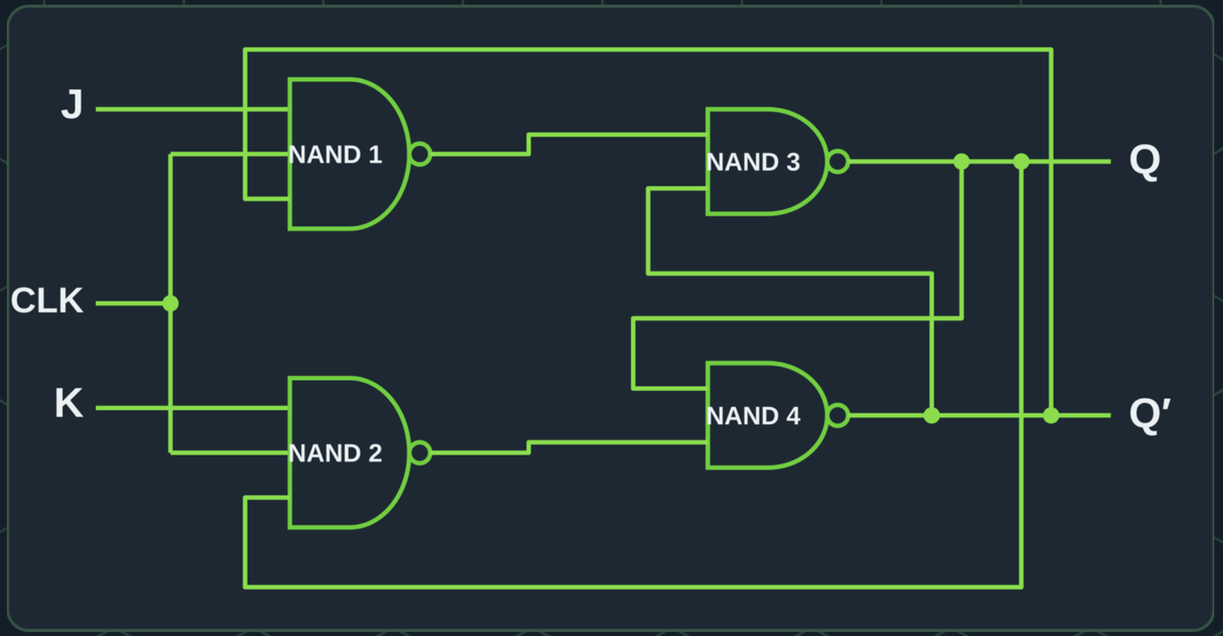
CLK	D	Q(n+1)	Operation
0	X	Qn	Hold
1 / ↑	0	0	Store 0
1 / ↑	1	1	Store 1

Characteristic equation

$$Q(n+1) = D$$

- When the clock is active, Q takes the value of D.
- When the clock is inactive, Q keeps its stored value.
- S = R = 1 can never occur, so there is no invalid state.

JK Flip-Flop Design



Q′ is fed back to the J gate and Q to the K gate. This feedback makes J = K = 1 toggle the output.

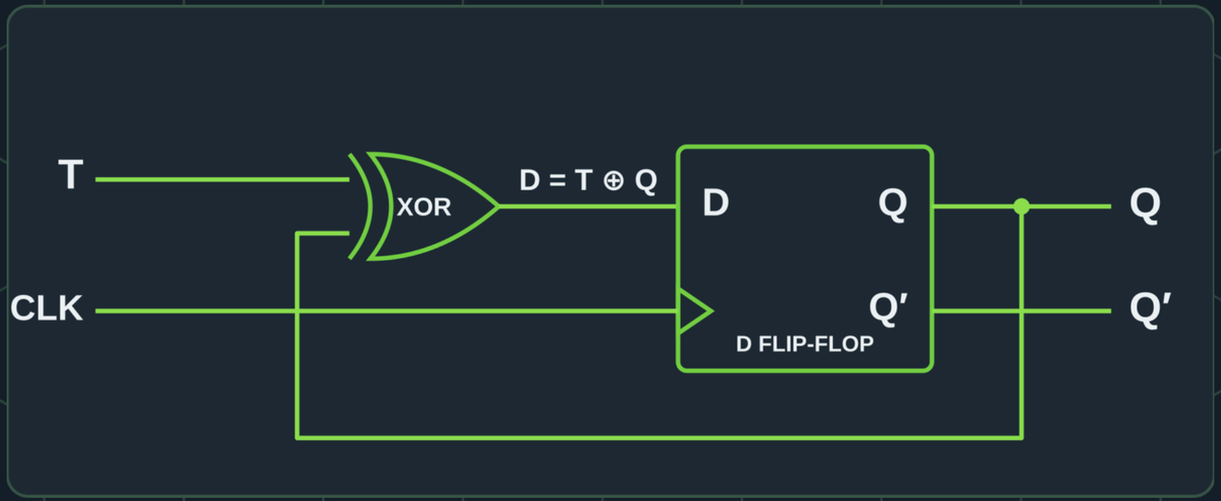
J	K	Q(n+1)	Operation
0	0	Qn	Hold
0	1	0	Reset
1	0	1	Set
1	1	Qn′	Toggle

Characteristic equation

$$Q(n+1) = J \cdot Q' + K' \cdot Q$$

With level triggering, J = K = 1 can toggle repeatedly while CLK is HIGH (race-around). Edge or master–slave triggering prevents this.

T Flip-Flop Design



T flip-flop = D flip-flop + XOR gate (D = T ⊕ Q). It is equivalent to a JK flip-flop with J = K = T.

T	Q(n+1)	Operation
0	Qn	Hold
1	Qn'	Toggle

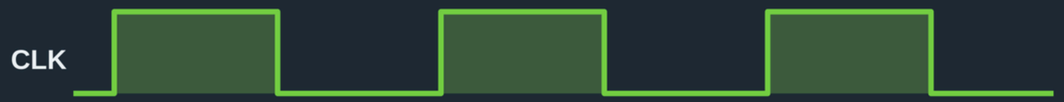
Characteristic equation

$$Q(n+1) = T \oplus Q = T \cdot Q' + T' \cdot Q$$

- T = 0: the output stays unchanged.
- T = 1: the output flips on every active clock edge.
- With T = 1 the output frequency is half the clock frequency.

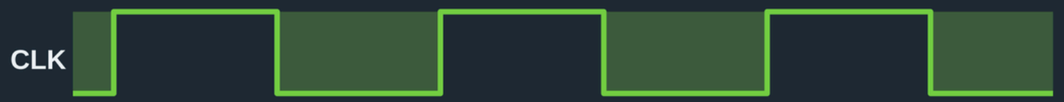
Level-Triggered vs Edge-Triggered Operation

HIGH-level triggering



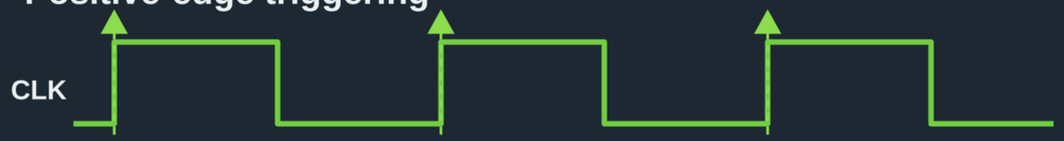
Active while CLK = 1 (shaded): output may change

LOW-level triggering



Active while CLK = 0 (shaded): output may change

Positive-edge triggering



Active only at the rising edge ↑ (0 → 1)

Negative-edge triggering



Active only at the falling edge ↓ (1 → 0)

Feature	Level-triggered	Edge-triggered
Output can change	While CLK is at its active level	Only at the active clock edge
Role of the clock	Enable (gate) signal	Sampling instant (trigger)
Input changes	Can reach the output while CLK is active	Ignored except at the edge
Typical device	SR latch, D latch	D, JK, T flip-flops

Comparison of SR, D, JK and T Circuits

Feature	SR latch	D flip-flop	JK flip-flop	T flip-flop
Inputs	S, R	D, CLK	J, K, CLK	T, CLK
Operation	Set, reset, hold (S = R = 1 invalid)	Stores D; no invalid state	Set, reset, hold, toggle	Hold, toggle
Triggering / control	Level: inputs act directly, no clock	Clock: level (latch) or edge	Clock edge (edge or master–slave)	Clock edge
Equation	$Q(n+1) = S + R'Q$	$Q(n+1) = D$	$Q(n+1) = JQ' + K'Q$	$Q(n+1) = T \oplus Q$
Applications	Switch debouncing, basic storage	Registers, data storage	Counters, shift registers, control logic	Counters, frequency dividers



Applications

Data storage

One flip-flop stores one bit; groups of them store data words.

Registers

n flip-flops form an n-bit register. Shift registers move data bit by bit.

Counters

T or JK flip-flops toggle to count pulses and divide frequency by 2.

Memory circuits

SRAM cells use cross-coupled gates, the same idea as the SR latch.

Digital systems

State machines, processors and control units use flip-flops to hold state.

Switch debouncing

An SR latch gives one clean transition from a bouncing switch.





Conclusion

- Latches and flip-flops are 1-bit memory elements that give circuits a stored state.
- SR latch is the basic cell. S = R = 1 is invalid.
- D flip-flop removes the invalid state and stores the data input.
- JK flip-flop supports set, reset, hold and toggle.
- T flip-flop holds or toggles, ideal for counters.
- Clock edge triggering keeps sequential systems synchronised and reliable.

SR latch

$$Q(n+1) = S + R'Q$$

D flip-flop

$$Q(n+1) = D$$

JK flip-flop

$$Q(n+1) = JQ' + K'Q$$

T flip-flop

$$Q(n+1) = T \oplus Q$$


Thank
You
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